INDEPENDENT PROOF OF CONCEPT / BUSINESS CASE STUDY

BeerIQ AI

Connected retail operations, from purchase to replenishment and reverse logistics

A discussion proposal inspired by first-hand retail experience at The Beer Store.



Figure 1. Executive overview — illustrative frame from the BeerIQ demonstration video; not a live application capture.

THE OPPORTUNITY

Explore whether a connected decision-support workflow could reduce repetitive handoffs across inventory, replenishment, warehouse requests and empty-container collections—without assuming gaps in existing company systems.

Executive summary

BeerIQ AI is an independently built Python and Streamlit prototype that joins simulated point-of-sale activity with inventory rules, ERP-style replenishment requests, warehouse and fleet coordination, and return-container collection logic. Its purpose is to make an operational concept tangible and measurable—not to claim an implemented solution at The Beer Store.

The proof of concept uses selected figures attributed to The Beer Store's published 2024 reports for historical context. Every transaction-level, store-level, inventory, route and ERP event displayed in the demo is synthetic. No live SAP, POS, customer or employee systems are connected. [1–3]

01 The workflow concept

The central idea is event-driven: routine actions become structured requests, while decisions with operational or financial consequences remain subject to human review.

Trigger	Proposed automated response	Control
Simulated customer sale	Deduct recorded units; assess inventory against a defined reorder threshold.	Reconcile stock and verify source data.
Low-stock event	Prepare an ERP-style replenishment request; show warehouse allocation status.	Approve order quantities and purchasing rules.
Fulfillment request	Create an illustrative delivery task and fleet assignment.	Dispatcher confirms vehicle, capacity and schedule.
Empty-container return	Update collection balance; suggest pickup when threshold reached.	Verify physical count and handoff.
Variance or failed step	Raise an exception with an audit-log entry.	Manager investigates; discrepancy is not evidence of theft.



Figure 2. ERP and replenishment sequence — illustrative video frame; requests are simulated, not transmitted to SAP.

Why integration could matter

Connecting these steps may improve visibility across teams and reduce duplicate data entry. The value depends on the current operational baseline, existing enterprise capabilities, integration complexity, staff adoption and exception volumes. The prototype does not establish that these functions are missing today.

02 Reverse logistics & operational oversight

Collection workflows deserve particular attention because return containers create their own storage, pickup, transport and reconciliation requirements.

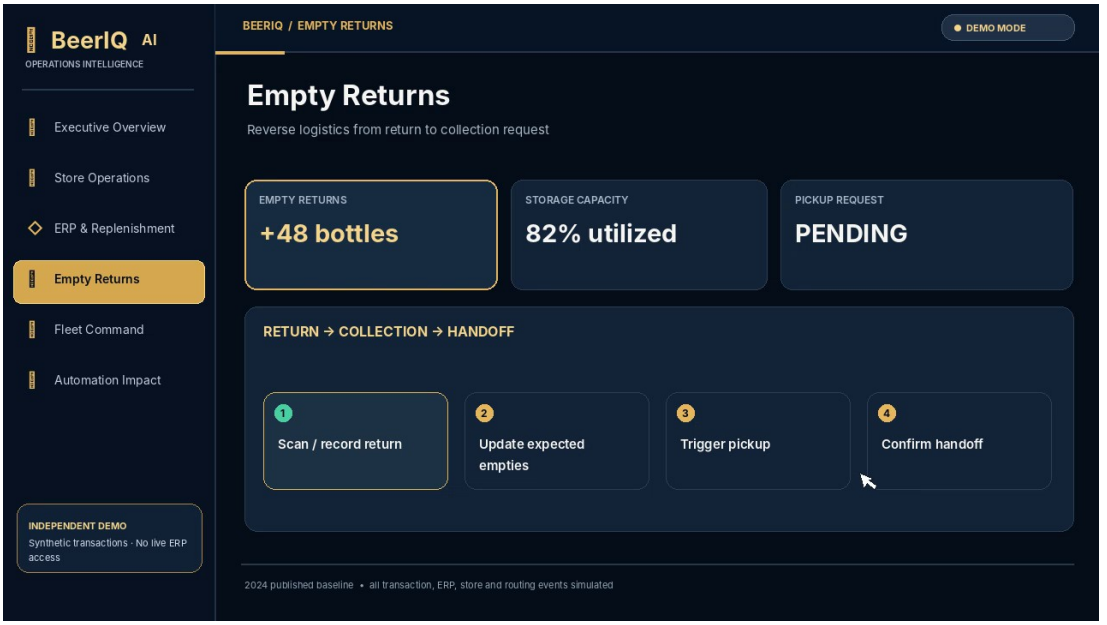


Figure 3. Empty-container collection logic — illustrative video frame, using fictitious return volumes and capacity values.

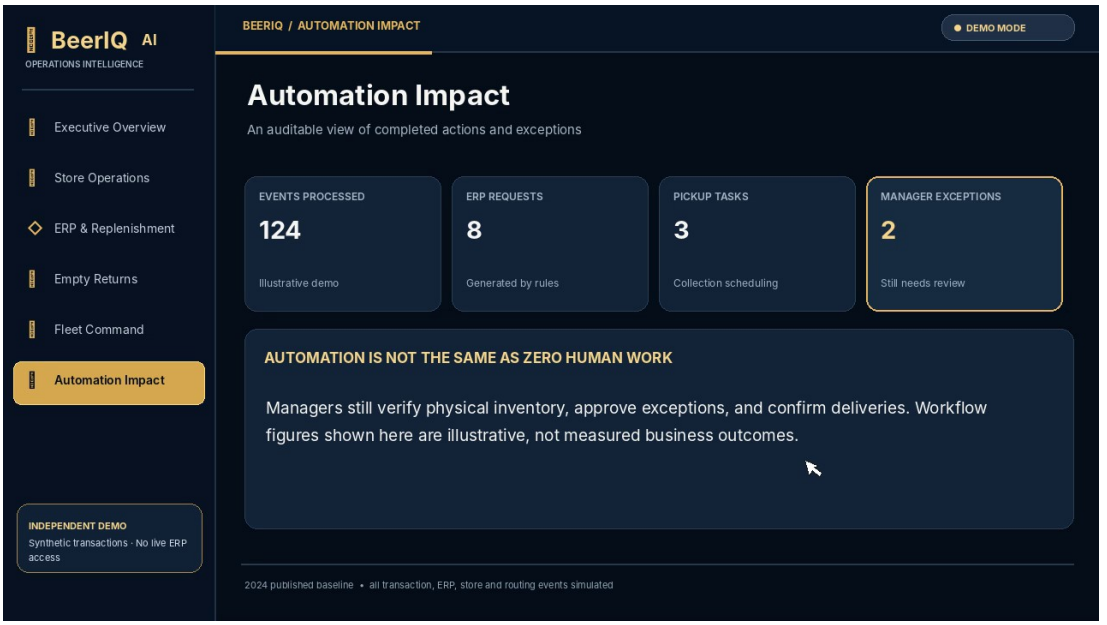


Figure 4. Automation impact view — illustrative video frame; event counts are simulation outputs, not measured business results.

Business outcomes to validate

Potential improvement	Pilot measurement
Reduce repetitive administration	Minutes per replenishment request and per pickup request.
Improve exception visibility	Open exceptions, time-to-resolution and false-alert rates.
Support inventory accuracy	Cycle-count variance, stockouts and order corrections.
Improve collection responsiveness	Overflow events, pickup lead time and verification effort.

03 Public reporting context

Published 2024 reporting helps illustrate scale and why the broader workflow is worth investigating. The numbers below are historical context, not data flowing through BeerIQ AI in real time.

Reported 2024 measure	Business relevance
3,145,896 hectolitres distributed [1]	Large-scale movement of product motivates a closer look at traceable operational handoffs.
407 retail stores and 26 distribution locations [1]	Multiple nodes create a meaningful environment for studying exception visibility.
More than 5,300 new retail customers reported [1]	Channel expansion makes process interoperability a useful discussion topic.
More than 1.6 billion containers collected across programs [2]	Reverse logistics is material enough to justify process-level experimentation.
76% recovery rate reported for the relevant program [2]	Any collection analysis should retain the report's precise metric definitions.

04 Proposed evaluation pilot

Phase	Approach	Decision evidence
Discovery	Map actual replenishment and returns processes with authorized stakeholders.	Current-state map; existing automation; baseline time studies.
Sandbox	Run synthetic or approved, de-identified events through three workflows.	Correctness, auditability and exception handling.
Shadow mode	Compare recommendations with the established workflow; take no automated action.	Agreement rates, false positives, staff effort.
Go / no-go	Review data access, security, integrations and user feedback.	Benefit-cost estimate and management sign-off.

Illustrative benefit framework

SCENARIO ONLY
100 stores × 20 minutes of potential administrative capacity per day × 300 days ÷ 60 = 10,000 hours per year. At an assumed CAD $25 per hour, that is CAD $250,000 of illustrative labour-capacity value—not demonstrated savings or a Beer Store forecast.

A credible business case must use observed time savings, account for implementation and maintenance costs, and distinguish staff capacity released from cash costs actually eliminated.

05 Scope, safeguards & next conversation

In scope today	Not established or connected
Python / Streamlit prototype; rules; synthetic event simulation; historical KPI visualization.	Actual Beer Store POS, SAP, warehouse management, dispatch or GPS integrations.
Demo audit trails and exception flags.	Production security, operational validation, savings or field deployment.
AI-generated / illustrated screen imagery used in demonstrations.	Official company screenshots, photographs, endorsement or brand approval.

Recommended next conversation

I would welcome feedback from retail operations, technology and supply-chain stakeholders on whether these workflows overlap with existing initiatives, which handoffs are most burdensome, and whether a narrowly scoped, non-production demonstration would be useful. The appropriate next step is discovery—not deployment.

VIEW THE PROTOTYPE
Interactive demo: https://kishoremathes18-byte-cmmt2kve6k4dcjjcew9stt.streamlit.app/
GitHub project: https://github.com/kishoremathes18-byte/BeerIQ-AI

Source notes & references

[1] Brewers Retail Inc. The Beer Store — 2024 Operations Report (2025). Public historical operating statistics and operational context. https://www.thebeerstore.ca/api/images/wyfly6ekgxs8/5J7jgki4GLkwyXquTRASlc/889686a44d43bc0218a01e1460ec0f0c/OpsReport2024_FINAL.pdf

[2] Brewers Retail Inc. The Beer Store — 2024 Stewardship Report (2025). Return-container collection and recovery reporting. https://www.thebeerstore.ca/api/images/wyfly6ekgxs8/gMafvCITsRs50ejsGtLG7/21fc02d4c93c5a48adf795566151bbe4/StewardshipReport2024_SinglePages.pdf

[3] BeerIQ AI — project source and documentation. https://github.com/kishoremathes18-byte/BeerIQ-AI

Figures 1–4 are stills from the project's narrated illustrative demonstration (BeerIQ_AI_Premium_60sec_Voiceover.mp4). They are not screenshots verified against the currently deployed application. Figures showing simulated quantities, orders or outcomes are not company results.

Independence and use disclaimer

BeerIQ AI is an independent educational and portfolio project inspired by retail experience. It is neither affiliated with nor endorsed by The Beer Store. The company's name is used for identification and attribution only. The project is not a replacement for approved company systems, and its benefits remain hypotheses pending evaluation.